AMNISH SINGH ARORA

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Software Developer with a strong background in **Web Development** and **AI engineering**, having relevant experience at both large enterprises and fast-paced startups.

Experience

Software Engineer – DeepStructure **Jul 2024 – Present**
San Francisco, CA

- Joined as the 6th employee at a Silicon Valley startup, developing a **TypeScript-based AI platform** that enables web developers to build and host agentic workflows powered by reactive data sources.
- Architected core components of our OpenAI's Assistants API re-implementation, including **streaming responses** (NATS), retries with **assertions**, **fallback models** with concurrently racing requests, robust **rate-limiting** using a TTL cache, and a comprehensive **smoke test suite** integrated into **CI**.
- Added support for using **non-OpenAI LLM providers** like Google, Anthropic, Sambanova, Cerebras, Groq, and Amazon Bedrock "through" OpenAI Assistants API.
- Implemented **dynamic self-improving** agents that leverage **user feedback** to generate increasingly relevant and fine-tuned responses over time.
- Reorganized the **structured completions** module in the **DeepStructure SDK** to follow best practices inspired by **open-source** projects like Vercel's AI SDK.
- Built an **LLM evaluation React dashboard** enabling side-by-side model comparisons using the "LLM as a Judge" approach and **Remix.js** for SSR.
- Optimized the performance and accuracy of structured completions in a **customer-facing** project by exploring various **prompt engineering** techniques.
- Built several reference apps for **customer demos,** leveraging Slack Interface and DeepStructure engine.
- Implemented a feature of **versioned Assistant families**, routing requests to a favorite within each family.

Frontend Developer – Ontario Treasury Board Secretariat **Sep 2022 – Jul 2024**
Toronto, ON

- Contributed to the **Angular** frontend of OnTrack, a large enterprise application integrating multiple legacy systems used by the Cabinet Office and Treasury Board, within an **Agile** team.
- Played a key role in a **major feature redesign**, enabling users to submit and apply **Suggestions** for Key Activities and Milestones tied to the Cabinet Office's election commitments.
- Developed features to generate document reports presenting **aggregated data** across multiple critical applications.
- Refactored **10** web applications and reusable packages to comply with WCAG 2.1 and WCAG 2.2 guidelines, ensuring **accessibility** for users with disabilities, resolving more than **200** issues.
- Helped **onboard new hires** including coop students and mid-level engineers.

Education

Seneca Polytechnic – Toronto (Affiliated to York University) **Sep 2021-Apr 2024**
Advanced Diploma: Computer Programming and Analysis

- Named to the President's Honor List in all semesters.
- Graduated with a perfect GPA of 4.0 and high honors.
- Opted for electives in Cloud Computing, Open Source, and iOS development.

Projects

Eslint-plugin-error-cause (URL):

- An ESLint plugin enforcing proper usage of the error cause property when re-throwing exceptions, reporting, and automatically fixing lost error context in **JavaScript** and **Typescript** projects.
- Efficiently traverses the JavaScript Abstract Syntax Tree to detect violating nodes and leverages modern best practices with fully type aware rules using typescript-eslint.
- Appreciated by the community at various platforms including Dev and Reddit, got accepted to the awesome-eslint open-source list, and a in progress to be integrated a new **core ESLint rule**.

Dine-A-Night (URL):

- A highly customizable restaurant **reservation** app, allowing restaurant owners to **visually design** seating layouts using a **drag-and-drop** interface**,** customers to reserve seats from the map, and admins to **monitor** upcoming and historical reservations.
- Built on top of a distributed architecture, with **Angular** Frontend, **Express.js** for business logic, **MongoDB** for structured data, **Firebase** for authentication and blob storage, and **Google Cloud** for Maps and Address Autocomplete features.
- Comes with a comprehensive User Manual and YouTube Demos for onboarding new users and attracting **investor interest**.

ChatCraft (URL)**:**

- 3rd largest contributor to a developer-oriented **AI chatbot** with **168** GitHub stars, featuring a function tools editor, speech capabilities, image processing, and more.
- Integrated **Text-to-Speech** capabilities like real-time audio streaming, customizable voice selection, and downloading messages as audio.
- Improved audio **download speeds** by **500%** by intelligently splitting message content and re-assembling concurrently generated audio chunks, also enabling downloads of infinitely long messages.
- Developed experimental features like Web Handlers, enabling users to seamlessly integrate and invoke their **backend microservices** directly within ChatCraft.

Technical Skills

- **AI Engineering:** Working knowledge of prompt engineering techniques like **RAG** and **Few-shot prompting**, and consuming/extending services offered by different AI providers.
- **Frontend Development:** Experience building highly interactive web apps leveraging modern Frontend frameworks like **Angular**, **React, Next.js,** and **Remix**. Deep understanding of semantic HTML and advanced CSS to build responsive and accessible designs.
- **Programming Languages:** Developed several projects in **C/C++**, **Python, Java, C#,** and **Swift**, written with clean, structured and maintainable code following best practices**.**
- **Backend Development:** Ability to build **secure** and reliable REST APIs and server-side rendered apps in Node frameworks like **Express.js** and **Hono**, and some experience in **ASP.NET.**
- **Cloud Computing:** Deep understanding of Cloud Fundamentals, **Docker,** and experience in consuming a wide range of services provided by major cloud providers like **AWS**, **Azure,** and **Google Cloud,** including authentication, storage, compute, and LLM infrastructure.
- **Databases:** Proficient in designing normalized relational database models and leveraging powerful features of multiple flavors of **SQL** like **Postgres, Oracle,** and **MySQL**. Familiar with tradeoffs between different types of **NoSQL** databases and experienced in using document databases like **MongoDB**, key-value stores like **DynamoDB**, and blob stores like **Amazon S3.**